

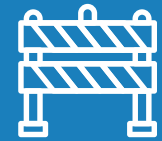
**SMART INDIA
HACKATHON
2026**

-
- The logo for the Smart India Hackathon (SIH) is centered on a large, light gray hexagon. Inside this hexagon is a stylized brain. The left half of the brain is orange and filled with a circuit board pattern. The right half is green and filled with white binary code (0s and 1s). Below the brain, the letters 'SIH' are written in a bold, dark gray, sans-serif font. The entire logo is surrounded by several smaller, light gray hexagons of varying sizes, some of which are partially cut off by the edges of the frame.

Problem Identification

THE 2-MINUTE OPD CRISIS

- Average consult lasts just 2.2 minutes (BMJ Open 2017) across 4,000–10,000 daily OPDs.
- 70–80% of diagnostic accuracy relies on history, leading to severe clinical errors.



The AYUSH Complexity Barrier

- Ayurvedic diagnosis requires deep Dashavidha Pariksha (Prakriti, Agni, Koshta).
- Impossible to capture manually in 120 seconds, forcing abbreviated holistic care.

Paper Chaos & Doctor Time-Drain

- Patients carry disordered, multilingual handwritten papers from multiple prior providers.
- Physicians waste 70% of consultation time reading physical files instead of examining.



The "First-Mile" ABDM Gap

- ABDM built national ABHA and FHIR rails, but lacks a gate intake tool at hospital entry.
- Existing kiosks strictly handle administrative tokens, completely ignoring clinical history.



ABDM & ABHA Ready



Deterministic
Red-Flag Interceptor



100% On-Premise
Privacy

PATIENT JOURNEY

1

Patient scans ABHA card or enters ABHA ID, Confirms with OTP.

2

Patient places old paper prescriptions and lab reports under the scanner.

3

Patient speaks symptoms naturally or taps picture options on screen.

4

System automatically checks for emergency warning signs in real time.

5

Kiosk poses quick follow-up questions to assess the illness and generate a medical summary.



Bharat-Ready
Voice Triage

01

Push-to-talk vernacular intake with noise suppression eliminates typing for patients.



Instant Paper-
to-EHR Digitizer

02

Clinical OCR digitizes prescriptions and lab reports into structured health data.



Evidence-
Grounded Dual
Medicine

03

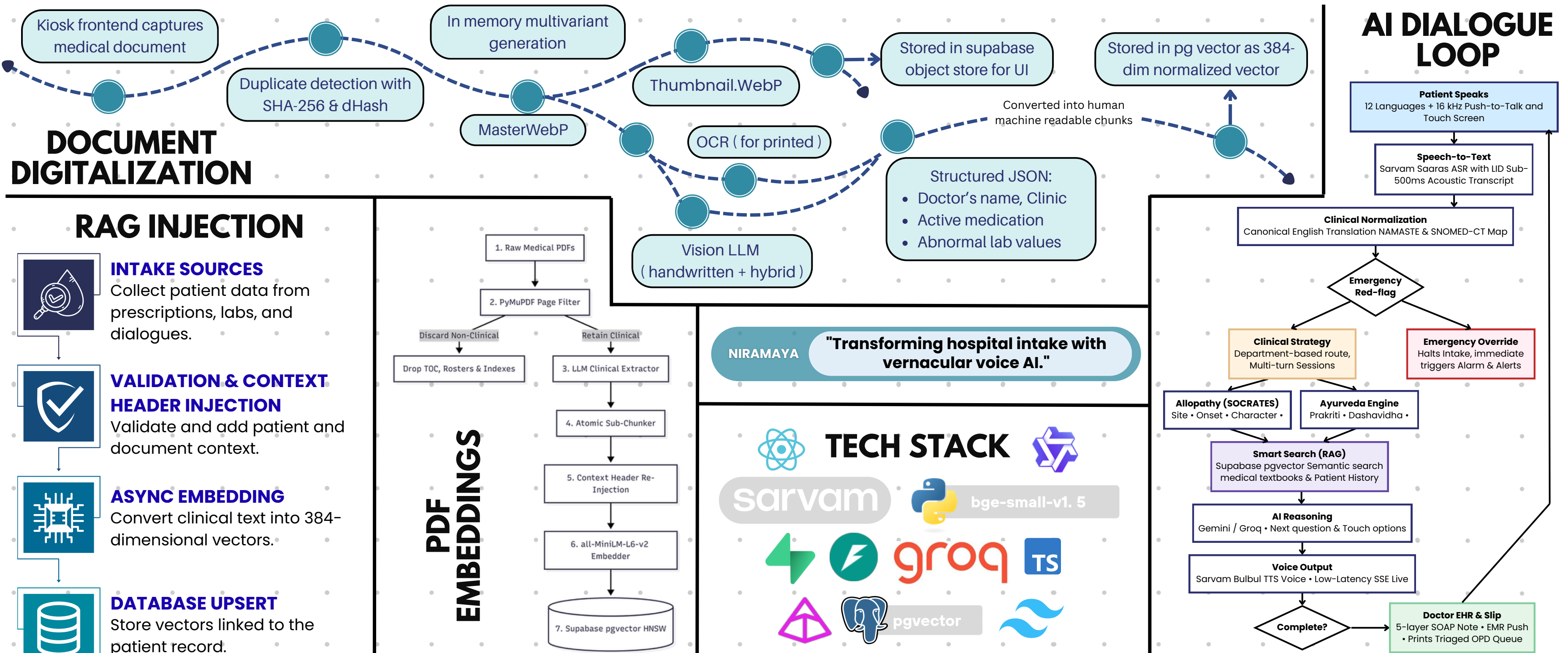
Evidence-grounded RAG combines ICMR protocols with standardized AYUSH assessments.



30-Sec Pre-
Consultation
Summary

04

Edge-computed clinical briefs reach doctors before patients enter consultation rooms.



Limitations of Existing Solution

Overwhelmed OPDs and Doctor Burnout:

High volume (4,000-10,000 patients/day) compresses visits to 2-3 mins. Doctors spend 60% of time gathering basic repetitive intake details rather than examination.

Language and Literacy Barriers :

Complicated English/Hindi-only portals alienate rural populations. Absence of native speech interface forces non-literate patients to skip medical history.

Lost Paper Records & High Repeat Costs:

40%+ of physical prescriptions are lost. Duplicate diagnostics cost hospitals ₹60L+/year. No automated ABDM/ABHA health-record linkage.

Feasibility of our Solution

Technical Feasibility:

Sub-10ms vector search powered by PostgreSQL pgvector (HNSW) & 384-dim bge-small-v1.5 embeddings. Dual ASR via Sarvam AI (Saaras) & Bhashini with offline faster-whisper fallback. Computer vision pre-ingestion gate validates prescription clarity in <1.2s.

Economic Feasibility:

Ultra-low operational cost of ₹0.02 to ₹0.05 per consultation via hybrid edge caching. Using zero-egress Cloudflare R2 / MeghRaj NIC intranet drops annual hospital record storage expense from ₹78 Lakhs (paper) → ₹1.57 Lakhs (~98% cost reduction).

Operational Feasibility:

Zero-learning-curve Push-to-Talk (PTT) voice interface with multilingual audio readback in 11 Indic languages. Deploys seamlessly on standard low-cost Android/Windows touch kiosks in District Hospitals, CHCs, and Ayushman Arogya Mandirs.

Potential Challenges and Risks

Ambient OPD Acoustic Noise:

High background clamour (crowd chatter, loud announcements, echoes at 70-85 dB) in government OPD halls causing ASR transcription drops or hallucinatory loops.

Unconstrained Document Quality:

Crumpled paper slips, motion-blurred smartphone scans, specular glare, and illegible doctor handwriting causing OCR errors or inaccurate clinical parsing.

Rural Network Volatility:

Intermittent 2G/3G connectivity or periodic internet blackouts in remote primary health sub-centres interrupting live cloud LLM API pipelines.

Strategies to overcome challenges

Dual-Stage DSP & ASR Safety Gate:

Client-side Web Audio spectral subtraction noise gate + RMS silence detection (<0.01) + repetitive hallucination loop guardrail with localized "Speak Again" prompts.

Computer Vision Pre-Ingestion Gate:

Real-time Laplacian variance blur rejection (score <100), specular glare mask filtering, perceptual dHash deduplication (<0.1ms), and 3-tier progressive WebP compression.

Offline-First Hybrid Edge Architecture:

Static clinical RAG cache (ICMR STW & CCRAS guidelines in `clinical_vectors_cache.json`), local quantized Whisper engine, and queued SQLite sync during outages.

BENEFICIARIES

Ministry of Ayush:

- Core Problem Solved: Unstandardized intake & misrouted patients
- Outcome: 100% NAMASTE/ICD-11 alignment; accurate 6-specialty Ayush routing



Health Ministry:

- Core Problem Solved: Massive OPD overcrowding & fragmented paper records
- Outcome: ABDM/FHIR integrated; real-time syndromic disease tracking



Patients:

- Core Problem Solved: Language barriers & 45-60 min waiting lines
- Outcome: Voice in regional languages; wait time reduced to < 2 minutes



Doctors:

- Core Problem Solved: Data-entry burnout & rushed consultations
- Outcome: 50% faster intake documentation; instant structured summary + allergy alerts



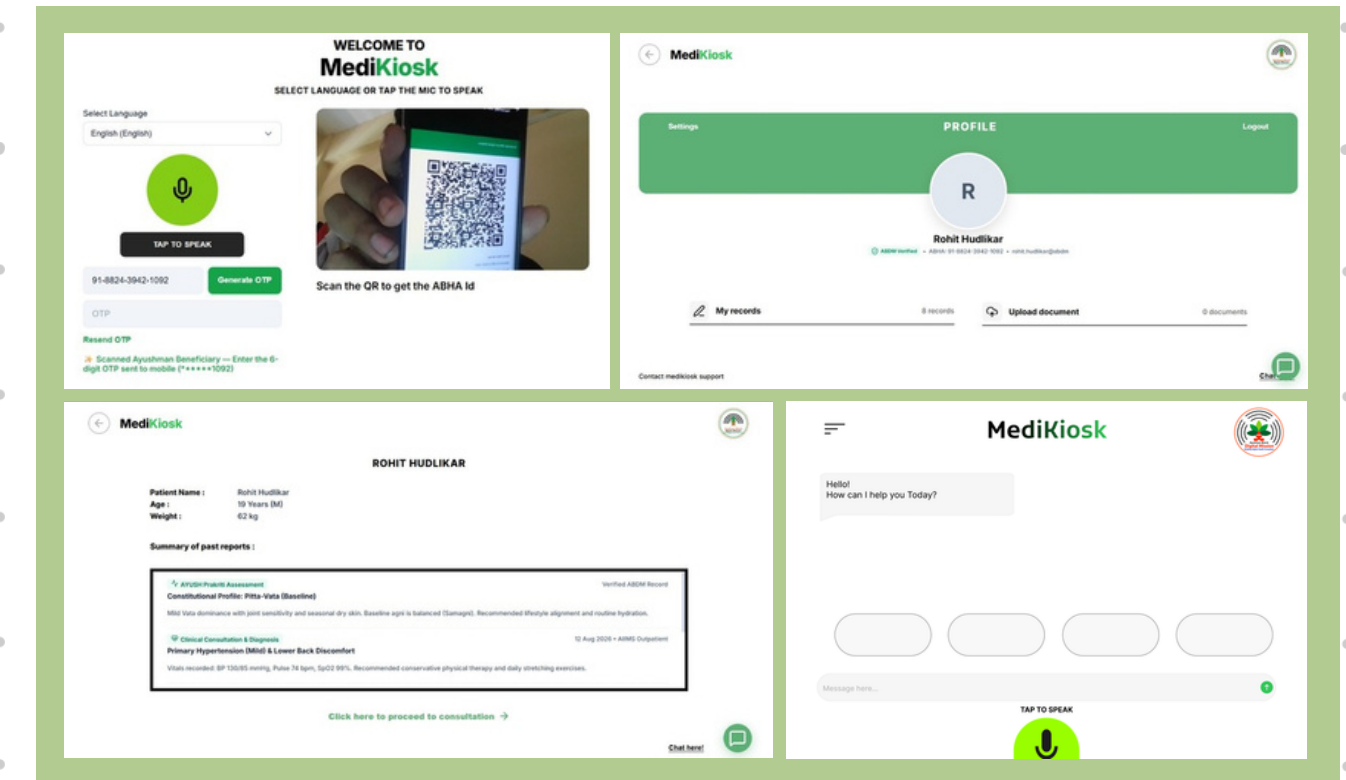
GitHub



PROTOTYPE:

Master Statistical Evaluation Report

Metric	Measured Score	What This Means Clinically
 Total Cases Evaluated	745	Substantial, statistically valid cohort of real-world patient symptom descriptions.
 Emergency Recall (Sensitivity)	97.54%	true emergencies and urgent cases were correctly caught by the LLM.
 Distractor Specificity	94.55%	routine/observation cases were correctly passed to normal OPD without panicking the patient.
 Positive Predictive Value (Precision)	93.55%	When the LLM flags a case as a Red Flag, it is genuinely an emergency in >93% of cases.
 F1-Score	95.42%	High harmonic balance between patient safety (recall) and avoiding hospital congestion (precision).
 Overall Triage Accuracy	95.87%	total correct triage classifications.



Literature Citations

- Dataset: Clinical Symptom Triage & Specialist Recommendation Dataset
- Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2)
- BioBERT: a pre-trained biomedical language representation model for biomedical text mining
- IndicTrans2: Towards High-Quality and Accessible Machine Translation for all 22 Scheduled Indian Languages
- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
- Ayurvedic Management of Common Disease Conditions – Treatment Protocols & Costing Guidelines, Volume I
- National Health Authority (NHA). Ayushman Bharat Digital Mission (ABDM) Handbook. Government of India.



RAG Citations

- Sentence Transformers Documentation – Pretrained Models
- Contextual Retrieval in AI Systems – Anthropic



MediKiosk is built strictly in accordance with national and international health informatics standards:

Allopathic References

- Standard Treatment Workflows (STW) of India: Volume 1 (2019)
- Standard Treatment Workflows (STW) of India: Volume 3 (2022)

Ayurveda References

- Dataset: NAMASTE Morbidity Master
- Development of a Standardized Prakriti Assessment Tool," multiple validation studies
- Review of Dashavidha Pariksha and Its Practical Applicability in Ayurveda



Digital Health & Interoperability Standards

- ABDM (Ayushman Bharat Digital Mission – NHA)
- HL7 FHIR Release 4
- Digital Personal Data Protection (DPDP) Act 2023 & MeitY

National Ayush & Clinical Standards

- CCRAS (Central Council for Research in Ayurvedic Sciences)
- NAMASTE Portal
- PCIM&H (Pharmacopoeia Commission for Indian Medicine & Homoeopathy)

Audio and OCR References

- Bhashini API Documentation
- Sarvam AI – API Documentation

